Exercise: Cyclic Code

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1. Generate (7,4) Cyclic code from the data (0101, 1001, 0010, 1110, 0000), and its hardware implementation.

cyclic code

0101

shifted 0101000

Divide by 1011 (modulo-2 division)

remain 100

code word 0101100

1001

shifted 1001000

Divide by 1011 (modulo-2 division)

remain 110

code word 1001110

0010

shifted 0010000

Divide by 1011 (modulo-2 division)

remain 110

code word 0010110

1110

shifted 1110000

Divide by 1011 (modulo-2 division)

remain 100

code word 1110100

0000

A message of all zero will always yield a remainder of 000

code word : 0000000

Hardware implement

The hardware for systematic cyclic encoding is implement using linear feedback shift Register

using 3 flip flop

2 XOR Gates

$g(x) = x^3 + x + 1$

